

Proposal for “Smart Healthcare Suite”

For

Phòng khám Gia ình Vit Úc
Vietnam Australia Family Health Services
(VAFHS)

Prepared & submitted by
IADCX | www.iadcx.com

Contents

1	Executive Summary	3
2	Introduction	3
3	Multicountry Deployment and Customization Framework	3
4	Solution Overview	4
4.1	Phase 1: ERP Module, Multilingual, Accounting, and Mobile Implementation	4
4.1.1	Solution Design for Phase 1	4
4.1.2	Patient Relationship Management	4
4.1.3	Appointment Scheduling & Telemedicine	5
4.1.4	Billing	5
4.1.5	EHR Integration via APIs	5
4.1.6	HR & Payroll	6
4.1.7	Inventory	6
4.1.8	Patient Portal	7
4.1.9	Field Appointment & Dispatch	7
4.1.10	Work Order Management	7
4.1.11	Mobile Access	8
4.1.12	Medical Equipment Maintenance	8
4.1.13	Home Health Services	8
4.1.14	SLA & Compliance	9
4.1.15	Reporting & Analytics	9
4.1.16	Accounting System	10
4.1.17	Patient Communication and Multilingual Support	10
4.1.18	Marketing Automation	10
4.1.19	Survey & Feedback	11
4.1.20	90-Day Project Implementation Plan for Phase 1	11
4.1.21	Pricing Options for Phase 1	12
5	Data Workflows	12
6	Non-Functional Requirements	12
6.1	Security & Compliance	12
6.2	Scalability & Reliability	13
6.3	Performance	13
6.4	User Experience	13
6.5	Language Support	13
7	Conclusion	13
A	Diagrams	13
A.1	Phase 1 Solution Diagram	13
A.2	Phase 1 Architecture Diagram	13

A.3 Data Flow Workflow..... 13

A.4 Solution Stack and Components..... 13

A.5 Gantt Chart for Phase 1 Implementation Plan..... 13

1 Executive Summary

This proposal presents a scalable healthcare system for Phòng khám Gia đình Vít Úc (VAFHS). Built on an open-source ERP platform, it integrates mobile applications, multilingual capabilities, and accounting systems. The solution streamlines clinical workflows, ensures HIPAA/VAFHS compliance, and supports multicountry deployment. Implementation targets completion within 3 months, enhancing clinical efficiency, patient engagement, and regulatory compliance across VAFHS's healthcare operations.

2 Introduction

Phòng khám Gia đình Vít Úc (VAFHS) aims to modernize healthcare through a unified platform addressing clinical, operational, and patient needs. This proposal details a single-phase implementation of a cloud-native healthcare suite, leveraging customizable ERP modules, accounting, and mobile applications. Features like telemedicine, remote monitoring, and multilingual support via SMS, WhatsApp, and Zalo cater to diverse healthcare ecosystems, with a customization framework enabling multicountry deployment, improving patient care and efficiency.

3 Multicountry Deployment and Customization Framework

The solution incorporates a flexible customization framework to support multicountry deployment, enabling VAFHS to adapt to diverse regulatory, linguistic, and operational requirements. Built on a modular, open-source ERP, the system allows region-specific configurations for compliance (e.g., HIPAA, GDPR, VAFHS), currency, tax systems, and languages (English, Vietnamese, others). Custom workflows, templates, and APIs ensure seamless integration with local healthcare systems, while scalable cloud infrastructure supports multi-region data replication. This framework ensures rapid adaptation to new markets, enhances accessibility for diverse populations, and maintains operational consistency, enabling VAFHS to expand efficiently across Southeast Asia and beyond.

The customization framework ensures that VAFHS can efficiently expand into new regions by adapting to local healthcare regulations, such as Vietnam's Ministry of Health standards or GDPR for European markets. The platform supports dynamic tax calculations for Vietnam's VAT system and can be configured for other tax regimes. Language support includes real-time translation APIs for patient communication, reducing barriers in multilingual settings. The cloud infrastructure, hosted on AWS or Google Cloud, leverages auto-scaling and multi-region replication to ensure high availability and low latency, critical for healthcare operations in diverse geographies. This approach minimizes deployment time and costs while maximizing reliability.

4 Solution Overview

The Smart Healthcare Suite is a cloud-native platform designed to unify healthcare management for VAFHS. It integrates a web-based ERP interface, mobile applications, and accounting systems. The implementation focuses on a single phase to ensure a structured rollout.

4.1 Phase 1: ERP Module, Multilingual, Accounting, and Mobile Implementation

This phase deploys customized ERP modules, multilingual capabilities, an accounting system, and mobile applications to establish a robust foundation for VAFHSs clinical, operational, and administrative workflows, ensuring accessibility and efficiency across Vietnams healthcare landscape.

4.1.1 Solution Design for Phase 1

- **Client Layer:** Web and mobile interfaces for clinicians, staff, field agents, and patients, adhering to Material Design and Human Interface Guidelines for usability.
- **Application Layer:** Open-source ERP modules for patient management, billing, scheduling, and inventory, with customizable workflows for VAFHSs healthcare context.
- **Data Layer:** Secure database with encrypted storage for financial data and operational logs, ensuring HIPAA/VAFHS compliance.
- **Communication Layer:** Multilingual notifications via SMS, WhatsApp, and Zalo, integrated with patient portal and mobile apps.
- **Customization Framework:** Configurable templates for region-specific compliance, tax systems, and multilingual interfaces.

4.1.2 Patient Relationship Management

Use Case: Manage patient interactions, histories, and personalized care plans to enhance engagement and improve healthcare delivery for VAFHS patients.

Features:

- Track patient interactions and medical histories in real-time.
- Manage targeted health campaigns for specific patient groups.
- Create personalized care plans based on patient data.
- Monitor engagement metrics via integrated dashboards.
- Automate follow-up reminders for appointments and treatments.

Benefits: Improves patient retention, enhances care quality, and streamlines engagement for better health outcomes and satisfaction.

4.1.3 Appointment Scheduling & Telemedicine

Use Case: Streamline in-person and virtual appointment management, enabling self-scheduling and telehealth for accessible, efficient patient care at VAFHS.

Features:

- Manage in-person and virtual appointments with automated confirmations.
- Integrate telehealth for secure virtual consultations.
- Send reminders via SMS, WhatsApp, and Zalo.
- Enable patient self-scheduling through web and mobile apps.
- Generate analytics reports for appointment trends and no-shows.

Benefits: Reduces no-shows, enhances access to care, and optimizes scheduling efficiency for clinicians and patients.

4.1.4 Billing

Use Case: Automate billing processes for OPD/IPD services, ensuring compliance with Vietnam's VAT and transparent financial management for VAFHS.

Features:

- Automate OPD/IPD billing with Vietnam VAT compliance.
- Generate detailed billing reports for financial transparency.
- Support multiple payment methods, including online payments.
- Automate invoice generation and track payment statuses.
- Integrate with insurance systems for seamless claims processing.

Benefits: Streamlines revenue cycle, ensures regulatory compliance, and enhances financial transparency for VAFHS operations.

4.1.5 EHR Integration via APIs

Use Case: Integrate with VAFHS's existing EHR system via APIs to enable seamless data exchange and enhance clinical workflows.

Features:

- Connect to existing EHR system using secure APIs.
- Synchronize patient records for real-time data access.

- Ensure HIPAA/VAFHS compliance with encrypted data exchange.
- Enable retrieval of medical histories and treatment data.
- Support structured data integration for analytics and reporting.

Benefits: Enhances interoperability, streamlines clinical workflows, and ensures secure, compliant access to patient data.

4.1.6 HR & Payroll

Use Case: Manage staff schedules, payroll, and compliance to streamline human resource operations and support VAFHSs administrative efficiency.

Features:

- Automate payroll calculations with tax compliance.
- Manage staff schedules and track attendance.
- Process leave requests via self-service portal.
- Generate payroll reports for audit purposes.
- Integrate with time-tracking for accurate records.

Benefits: Reduces administrative workload, ensures labor law compliance, and improves staff management efficiency.

4.1.7 Inventory

Use Case: Track and manage medical supplies and equipment to ensure availability and compliance for VAFHSs clinical operations.

Features:

- Monitor stock levels in real-time with alerts.
- Automate inventory replenishment for critical supplies.
- Track expiration dates and reduce wastage.
- Integrate with billing for cost tracking.
- Support multi-site inventory management.

Benefits: Ensures supply availability, reduces waste, and enhances cost efficiency in clinical operations.

4.1.8 Patient Portal

Use Case: Provide patients with a self-service platform for appointments, medical records, and communication to enhance engagement and autonomy at VAFHS.

Features:

- Enable self-service appointment booking and rescheduling.
- Provide secure access to medical records.
- Support multilingual interfaces for diverse patients.
- Allow online bill payments and secure messaging.
- Integrate with telehealth for virtual consultations.

Benefits: Empowers patients, improves accessibility, and reduces administrative burden for enhanced care delivery.

4.1.9 Field Appointment & Dispatch

Use Case: Assign and track field tasks for agents to support home health services and improve operational efficiency for VAFHS.

Features:

- Assign tasks to field agents with real-time updates.
- Enable GPS navigation for optimized agent routes.
- Track task completion and generate reports.
- Integrate with mobile apps for seamless updates.
- Support task prioritization for urgent needs.

Benefits: Enhances field operation efficiency, improves response times, and supports scalable home healthcare delivery.

4.1.10 Work Order Management

Use Case: Manage service and maintenance orders to ensure timely equipment servicing and operational reliability for VAFHS facilities.

Features:

- Track and assign service orders automatically.
- Monitor work order progress in real-time.
- Generate reports for service order analytics.
- Integrate with inventory for spare parts tracking.

- Support priority-based scheduling for critical tasks.

Benefits: Ensures equipment reliability, reduces downtime, and streamlines maintenance processes for VAFHS.

4.1.11 Mobile Access

Use Case: Provide field agents and clinicians with mobile access to real-time data for efficient task management and patient care at VAFHS.

Features:

- Enable real-time data access via mobile apps.
- Support offline functionality for remote areas.
- Integrate with EHR for patient data updates.
- Provide GPS navigation for field agents.
- Support multilingual interfaces for accessibility.

Benefits: Enhances mobility, improves field efficiency, and ensures secure access to critical data.

4.1.12 Medical Equipment Maintenance

Use Case: Schedule and track equipment maintenance to minimize downtime and ensure compliance for VAFHSs operational reliability.

Features:

- Schedule routine equipment servicing with alerts.
- Track maintenance histories for compliance.
- Integrate with inventory for parts management.
- Generate maintenance reports for audits.
- Support predictive maintenance analytics.

Benefits: Reduces equipment downtime, ensures compliance, and enhances reliability of clinical operations.

4.1.13 Home Health Services

Use Case: Support remote patient monitoring and virtual care to improve accessibility and outcomes for VAFHSs home health patients.

Features:

- Integrate wearable device data for monitoring.

- Enable virtual consultations with clinicians.
- Track health metrics and generate reports.
- Support caregiver coordination via mobile apps.
- Provide patient education content securely.

Benefits: Enhances remote care, improves patient outcomes, and increases accessibility for underserved areas.

4.1.14 SLA & Compliance

Use Case: Ensure regulatory compliance and secure operations to maintain data integrity and trust in VAFHSs healthcare system.

Features:

- Implement HIPAA-compliant encryption and authentication.
- Maintain audit logs for regulatory compliance.
- Support role-based access control for security.
- Generate compliance reports for audits.
- Monitor data breaches with alerts.

Benefits: Ensures regulatory compliance, protects patient data, and maintains trust in VAFHS operations.

4.1.15 Reporting & Analytics

Use Case: Provide real-time dashboards and analytics to optimize clinical and operational decision-making for VAFHS management.

Features:

- Generate real-time KPI dashboards for operations.
- Provide analytics for resource planning.
- Track clinical and operational performance metrics.
- Support customizable report templates.
- Integrate with EHR for data-driven insights.

Benefits: Enhances decision-making, optimizes resources, and improves clinical and operational efficiency.

4.1.16 Accounting System

Use Case: Automate financial transactions and ensure tax compliance to streamline VAFHSs financial operations and reporting processes.

Features:

- Automate accounts payable and receivable processes.
- Generate financial statements for audits.
- Ensure compliance with Vietnam tax laws.
- Support multi-currency transactions for flexibility.
- Integrate with billing for financial tracking.

Benefits: Streamlines financial operations, ensures tax compliance, and enhances transparency for VAFHS.

4.1.17 Patient Communication and Multilingual Support

Use Case: Enable multilingual patient notifications and two-way communication to enhance engagement and accessibility for VAFHSs diverse patient base.

Features:

- Send multilingual notifications via SMS/WhatsApp/Zalo.
- Support two-way communication for patient inquiries.
- Automate appointment and health tip reminders.
- Track communication history and delivery status.
- Analyze language usage for improved engagement.

Benefits: Improves patient engagement, enhances accessibility, and reduces communication barriers for diverse populations.

4.1.18 Marketing Automation

Use Case: Run targeted health campaigns to promote services and engage communities, increasing awareness and participation in VAFHS programs.

Features:

- Automate email and SMS health campaigns.
- Track campaign performance with analytics.
- Segment audiences for targeted outreach.
- Promote preventive care and services.

- Integrate with social media platforms.

Benefits: Increases service awareness, boosts community engagement, and enhances preventive care outreach.

4.1.19 Survey & Feedback

Use Case: Collect and analyze patient feedback to improve service quality and patient satisfaction across VAFHSs healthcare services.

Features:

- Automate patient satisfaction survey collection.
- Support anonymous feedback for honest responses.
- Analyze feedback for service improvements.
- Generate summary reports for quality metrics.
- Integrate with patient portal for accessibility.

Benefits: Enhances service quality, improves patient satisfaction, and drives continuous improvement in care delivery.

4.1.20 90-Day Project Implementation Plan for Phase 1

- **Days 1-15: Requirements and Planning**
 - Conduct stakeholder workshops to gather requirements
 - Finalize functional and non-functional specifications
 - Define customization needs for VAFHSs context
 - Set up project management tools and timelines
- **Days 16-30: System Setup and Configuration**
 - Deploy cloud-based ERP infrastructure
 - Configure multilingual settings for English and Vietnamese
 - Set up accounting system with Vietnam tax compliance
 - Establish mobile app framework for iOS and Android
- **Days 31-60: Module Development and Integration**
 - Implement ERP modules (Patient, Billing, EHR, etc.)
 - Integrate mobile apps with ERP and notification systems
 - Configure SMS/WhatsApp/Zalo communication channels
 - Perform unit and integration testing for modules

- **Days 61-90: Testing and Deployment**
 - Conduct user acceptance testing with VAFHS staff
 - Train clinicians, staff, and field agents
 - Perform system performance and security testing
 - Deploy Phase 1 to production environment

4.1.21 Pricing Options for Phase 1

Pricing includes development, implementation, training, and hosting, offered in two models:

- **Term Plan (Yearly Term):**
 - Subscription: \$19,200/year (includes software, hosting, maintenance)
 - Includes: Application updates and L1 support
 - Training: \$3,000 (online, 2 English batches at \$1,000 + 1 Vietnamese at \$1,000)
- **Perpetual Plan:**
 - License Cost: \$45,000
 - Includes: Updates and L1 support
 - Training: \$3,000 (online, as above)
 - Hosting: At Amazon datacenter charges paid by VAFHS
 - AMC: 12% of perpetual license value

5 Data Workflows

- **Remote Monitoring:** Processes wearable device data for real-time health metrics and dashboards, integrated with patient portal for VAFHS clinicians.

6 Non-Functional Requirements

6.1 Security & Compliance

- Ensure HIPAA-compliant encryption and authentication
- Implement role-based access control
- Maintain audit logging

6.2 Scalability & Reliability

- Support auto-scaling architecture
- Ensure high availability with replication
- Optimize performance with load balancing

6.3 Performance

- Optimize API response times for EHR integration

6.4 User Experience

- Provide intuitive interfaces
- Adhere to Material Design and Human Interface Guidelines

6.5 Language Support

- Enable multilingual notifications via SMS, WhatsApp, and Zalo

7 Conclusion

The Smart Healthcare Suite equips VAFHS with a modern platform to enhance healthcare delivery in Vietnam and beyond. Through a single-phase implementation, it integrates open-source ERP, multilingual support, accounting, and mobile apps, with pricing options of \$19,200/year (Term Plan) or \$45,000 (Perpetual Plan), ensuring efficiency, compliance, and superior patient care within a 3-month timeline.

A Diagrams

A.1 Phase 1 Solution Diagram

A.2 Phase 1 Architecture Diagram

A.3 Data Flow Workflow

A.4 Solution Stack and Components

A.5 Gantt Chart for Phase 1 Implementation Plan

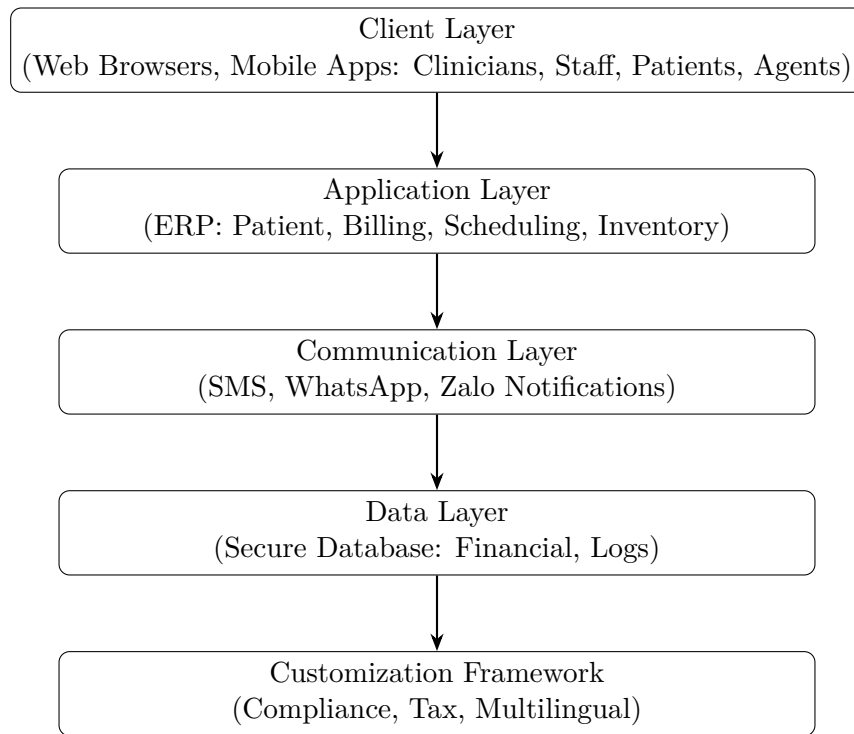


Figure 1: Phase 1 Solution Diagram

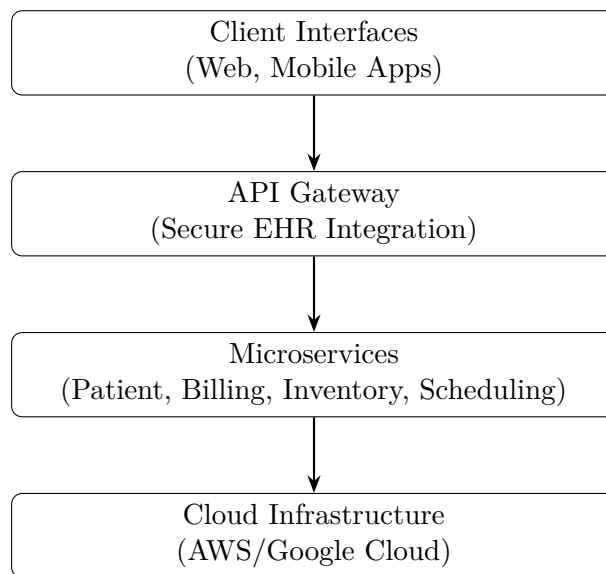


Figure 2: Phase 1 Architecture Diagram

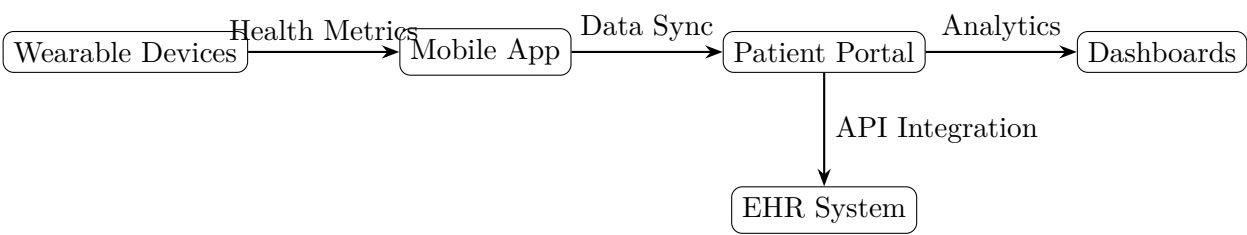


Figure 3: Data Flow Workflow for Remote Monitoring

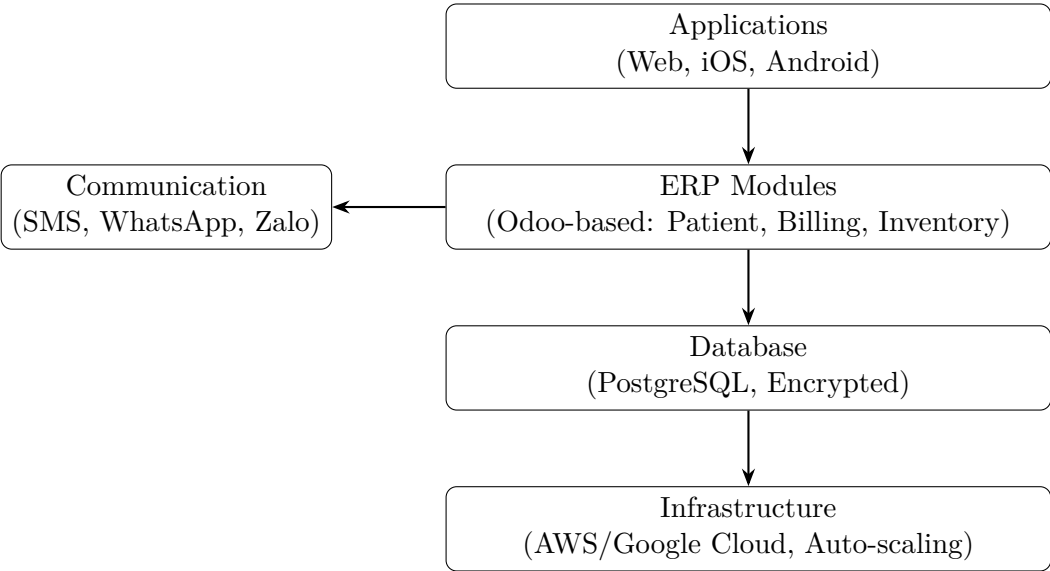


Figure 4: Solution Stack and Components

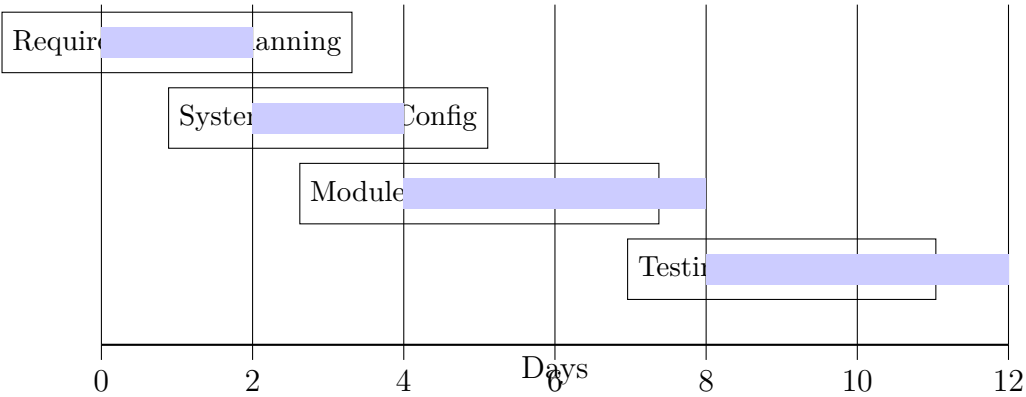


Figure 5: Gantt Chart for Phase 1 Implementation Plan